

- 1 A student is creating a presentation containing images and sound.

One of the files is an audio recording.

Complete the description of how a computer stores sound. Fill in the gaps using the given list of terms. Not all terms will be used.

analogue	colour depth	higher	repeated	smaller
binary	digital	lower	sample rate	unique
bit depth	height	measuring	sampling	
byte	hertz	recording	seconds	

An sound wave needs to be converted into a digital sound wave.

Sound is when the amplitude of the sound wave is measured at set intervals.

The is the number of times a second the sound wave is measured. This is given in Hertz.

Each amplitude is given a binary number. The number of bits allocated to each sample is the

The the number of bits, the wider the number of amplitudes can be measured.

[6]

i. Tick (✓) **one** box to identify the correct description of sound sampling.

☐

The frequency of the wave is measured a set number of times each second.

☐

The amplitude of the wave is measured at set intervals.

☐

The digital sound wave is measured a set number of times each second.

☐

The analogue sound wave's resolution is measured at set intervals.

[1]

ii. Explain how changing the bit depth will affect the sound file.

[2]

ii. Explain what is meant by a sample rate of 48 KHz.

----- [2]

iii. Describe how the file size of the sound recording could be reduced.

----- [4]

5 Layla is an artist. She draws images by hand. The image is then scanned and stored on a computer.

Layla stores her images on a secondary storage device.

i. Each image has a fixed size of 1 MB. The storage device has a capacity of 3 GB.

Calculate how many images can be saved on the storage device. Show your working.

.....
....._images

[2]

- ii. Layla uses the images to make videos. These videos are stored on her computer's internal storage device.

Identify the most appropriate type of storage device for Layla to use in her computer. Justify your choice.

Type of storage device

Justification

.....
.....
.....
.....

[3]

- iii. The videos include sound. The table has **four** statements about the storage of sound in a computer.

Tick (✓) **one** box in each row to identify if the statement is true or false

	True	False
The sample rate is the number of times the amplitude is recorded per second		
The smaller the bit depth the smaller the range of sounds recorded		
The larger the sample rate the larger the bit depth		
The frequency and pitch of the sound wave are measured		
Sound is stored using pixels		

[3]

- 6 A radio station records an interview with a computer scientist using a computer and audio recording software.

- i. Explain how sampling is used to store audio recordings.

[2]

- ii. A second interview with the computer scientist is recorded. Before this interview, the sampling frequency in the audio software is increased.

Define what is meant by the term **sampling frequency**.

[1]

- iii. Tick (✓) **two** boxes to show the effects of increasing the sampling frequency.

Data type of returned value	Tick (✓) two boxes
The file size of the digital recording will be smaller.	
The file size of the digital recording will be larger.	
The quality of playback of the digital recording will be better.	
The quality of playback of the digital recording will be worse.	

[2]

7 A computer records an audio file of someone playing a guitar.

Describe what happens when the computer converts the music into a file.

----- [2]

8 Alex needs to create an audio recording of himself playing his guitar.

i. Explain how sampling is used to make the recording.

----- [3]

ii. State the effects of increasing the sample rate of the recording.

----- [2]

9(a) Kofi uses his computer to record an audio file of himself playing his guitar.

Outline what happens when the computer converts the music into a file.

----- [2]

(b) Kofi increases the sample rate his computer is using to record his guitar.

Explain **two** effects this will have on Kofi's recording.

1

2

[4]

10 When recording a sound file on a computer, the sound needs to be sampled.

i. Describe how sampling is used when storing sound.

[2]

ii. Explain the effect of the sampling interval on the size and quality of the sound file recorded.

[3]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1			1 mark for each completed term to max 6 An analogue sound wave needs to be converted into a digital sound wave. Sound sampling is when the amplitude of the sound wave is measured at set intervals. The sample rate is the number of times a second the sound wave is measured. This is given in Hertz. Each amplitude is given a unique binary number. The number of bits allocated to each sample is the bit depth. The higher the number of bits, the wider the number of amplitudes can be measured.	6	<u>Examiner's Comments</u> Candidates were often able to demonstrate a good understanding of sound representation, identifying the appropriate missing terms. Candidates most commonly correctly identified that an analogue wave is converted into digital. The final space was often inaccurately completed with smaller, when instead it is the higher the number of bits the wider the number of amplitudes.
			Total	6	
2		i	1 mark for The amplitude of the wave is measured at set intervals	1	2+ ticks = 0 marks <u>Examiner's Comments</u> Some candidates were able to correctly identify that it was the amplitude that is measured at set intervals. A common error was that the frequency of the wave is measured, the frequency is a technical sound term that relates to the pitch of the wave, or the number of times the wave changes.

Question			Answer/Indicative content	Marks	Guidance
		ii	1 mark each to max 2 • The number of bits per sample will change // by example e.g. there will be more/less bits per sample • The file size will change // by example e.g. the file size will increase/decrease • There will be a change in the accuracy of each sample/amplitude/sound // by example e.g. more precise amplitudes // by example e.g. a wider/smaller range of amplitudes can be recorded • The quality will change // there will a different amount of distortion // by example e.g. the quality will improve/decline	2	MP3 needs to be clearly a wider range of amplitudes can be recorded i.e. more different values. Not that there are more amplitudes/samples per second. MP3 – 'more amplitudes can be measured' is BOD, but 'more amplitudes measured per second' is incorrect. BOD 'sound' for 'amplitude' e.g. for MP3 "a larger range of sounds can be recorded." <u>Examiner's Comments</u> This question was answered well by many candidates who were most commonly able to identify that the file size would change. Many candidates gave this through an example that when the bit depth increases the file size also increases. Candidates also often identified that the quality of the sound would increase, or that the sound would become more accurate when compared to the sound being recorded. Some candidates incorrectly identified that the bit depth would result in more samples being taken per second.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance																				
3		i	1 mark per bullet to max 3 • (analogue) sound wave is sampled • ... amplitude/height (of wave) is measured • ... at set/regular time intervals // by example • Each sample/measurement is stored as a binary number • The binary number for each sample is stored sequentially	3	MP2 do not award frequency of the wave is measured <u>Examiner's Comments</u> This question required candidates to describe how an analogue sound wave is measured and converted into digital. Many candidates were able to identify that the sound wave is sampled, with some also identifying that it was the amplitude (or wave height) that is sampled.  Misconception A common misunderstanding is that the wave frequency is used. The frequency in sound sampling is the number of samples taken each second or in sound waves is the number of times the wave has high and low amplitudes within a given time. It is not the number of changes in a second that is recording, it is the height of the wave.																				
		ii	1 mark for each row <table><tr><th>Change</th><th>File size increases</th><th>File size decreases</th><th>Accuracy increases</th><th>Accuracy decreases</th></tr><tr><td>Duration changes from 10 minutes to 20 minutes</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>Sample rate changes from 44 kilohertz to 8 kilohertz</td><td></td><td>✓</td><td></td><td>✓</td></tr><tr><td>Bit depth changes from 8 bits to 16 bits</td><td>✓</td><td></td><td>✓</td><td></td></tr></table>	Change	File size increases	File size decreases	Accuracy increases	Accuracy decreases	Duration changes from 10 minutes to 20 minutes	✓				Sample rate changes from 44 kilohertz to 8 kilohertz		✓		✓	Bit depth changes from 8 bits to 16 bits	✓		✓		3	<u>Examiner's Comments</u> Many candidates were able to correctly identify that the increase in duration increases the file size. The additional two changes were also often given accurately, but some candidates only gave one tick on each row for these, for example correctly identifying the change in file size but not accuracy.
Change	File size increases	File size decreases	Accuracy increases	Accuracy decreases																					
Duration changes from 10 minutes to 20 minutes	✓																								
Sample rate changes from 44 kilohertz to 8 kilohertz		✓		✓																					
Bit depth changes from 8 bits to 16 bits	✓		✓																						
			Total	6																					
4		i	• The height / amplitude... • ...as a numerical value • ...of the wave(form)	2	DO NOT accept frequency Do not accept "in binary" (given in question)																				

Question			Answer/Indicative content	Marks	Guidance
		ii	• 48,000 samples taken... • ...per second	2	BOD How often samples are taken // frequency of samples
		iii	e.g. • Reduce the sample rate (from 48KHz to a lower rate) • ...so fewer samples are taken per second • Reduce the bit depth (from 24 bits to a lower bit depth) • ...so less data is used for each sample • Use lossy compression... • ... to remove data (that won't be noticed) • Use lossless compression... • ...to identify patterns in the data • ...store this more efficiently • Reduce the length of the sound file ... • ... by example (from 30 seconds to a lower length) // less data to store	4	Any 4 points for 1 mark each Allow "compression" by itself for 1 mark if no other compression mark awarded. Allow suitable expansion of this for 1 mark. Do not accept "data is not lost" as expansion for lossless or "data is lost" as expansion for lossy.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance																		
5		i	1 mark for working, 1 mark for answer 1000*3 = 3000 images	2																			
		ii	1 mark for suitable type i.e. solid state // magnetic 1 mark per bullet to justification to max 2 solid state e.g.: • Large enough capacity • Can move computer without damaging storage • Faster access speeds magnetic e.g.: • Largest capacity • Do not need to move computer so moving parts do not matter • More reliable long-term	3																			
		iii	1 mark for 2 correct ticks 2 marks for all 3 or 4 correct ticks 3 marks for all correct <table border="1"><thead><tr><th></th><th>True</th><th>False</th></tr></thead><tbody><tr><td>The sample rate is the number of times the amplitude is recorded per second</td><td>✓</td><td></td></tr><tr><td>The smaller the bit depth the smaller the range of sounds recorded</td><td>✓</td><td></td></tr><tr><td>The larger the sample rate the larger the bit depth</td><td></td><td>✓</td></tr><tr><td>The frequency and pitch of the sound wave are measured</td><td></td><td>✓</td></tr><tr><td>Sound is stored using pixels</td><td></td><td>✓</td></tr></tbody></table>		True	False	The sample rate is the number of times the amplitude is recorded per second	✓		The smaller the bit depth the smaller the range of sounds recorded	✓		The larger the sample rate the larger the bit depth		✓	The frequency and pitch of the sound wave are measured		✓	Sound is stored using pixels		✓	3	2 ticks in 1 row is incorrect
	True	False																					
The sample rate is the number of times the amplitude is recorded per second	✓																						
The smaller the bit depth the smaller the range of sounds recorded	✓																						
The larger the sample rate the larger the bit depth		✓																					
The frequency and pitch of the sound wave are measured		✓																					
Sound is stored using pixels		✓																					
			Total	8																			

Question			Answer/Indicative content	Marks	Guidance										
6		i	1 mark per bullet to max 2. • Height/amplitude of waveform is sampled/measured • Converted to / stored as binary/digital • Sample / measurements taken at a regular interval / set interval / by sensible example (eg 44,000 times per second)	2 AO1 1b (2)	Do not accept frequency Do not accept unrealistic sample rates (e.g. once per second). <u>Examiner's Comments</u> Candidates generally understood that sampling involves taking measurements at regular or set intervals, and that these measurements are then converted to and stored as binary. Higher ability candidates were able to state that it is the height or amplitude of the waveform that is measured. Some candidates were confused by this and incorrectly tried to explain sampling as a measurement of frequency.										
		ii	1 mark per bullet to max 1. • number of samples taken per second / per time period • How <u>often/regularly</u> a sample is taken	1 AO1 1a (1)	Accept reference to Hertz (Hz) as time period. <u>Examiner's Comments</u> Most candidates were able to identify sample frequency as how often a sample is taken. A slightly more precise answer was that it is the number of samples taken per time-period, such as 44,000 samples per second which can be given as 44KHz (1 Hertz = once per second).										
		iii	1 mark per tick to max 2. <table border="1"><tr><td></td><td>Tick (✓) two boxes</td></tr><tr><td>The file size of the digital recording will be smaller</td><td></td></tr><tr><td>The file size of the digital recording will be larger</td><td>✓</td></tr><tr><td>The quality of playback of the digital recording will be better.</td><td>✓</td></tr><tr><td>The quality of playback of the digital recording will be worse.</td><td></td></tr></table>		Tick (✓) two boxes	The file size of the digital recording will be smaller		The file size of the digital recording will be larger	✓	The quality of playback of the digital recording will be better.	✓	The quality of playback of the digital recording will be worse.		2 AO1 1b (2)	If 3 ticks given, max 1 mark If 4 ticks given, 0 marks.
	Tick (✓) two boxes														
The file size of the digital recording will be smaller															
The file size of the digital recording will be larger	✓														
The quality of playback of the digital recording will be better.	✓														
The quality of playback of the digital recording will be worse.															

Question			Answer/Indicative content	Marks	Guidance
			Total	5	
7			- The height of the wave is measured/sampled (at regular/set intervals) - Turned into/stored as binary	2 (AO1 1b)	1 mark for each bullet, to a maximum of 2.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance
8		i	• The amplitude / height of the wave is measured • At set / regular intervals // by reasonable example • And stored as a binary number • The samples form an approximated sound wave	3	NOT frequency / pitch NB For the second bullet, this must relate to set intervals / the same interval. A set number of times per second does not suggest the same intervals. Examiner's Comments Where candidates knew how sound was stored, they were able to give a clear description, and those better candidates did well with this question. Some candidates did not understand the terminology of sampling, and often guessed at the meaning of the word.
		ii	• File size increases • So the sound is truer / better quality / more accurate compared to the original / analogue	2	Examiner's Comments Many candidates were able to correctly identify that the file size increased. Fewer candidates could express that the quality was improved because it was closer to the original that was being recorded.
			Total	5	
9	a		• The height of the wave is measured / sampled (at regular / set intervals) • Turned into / stored as binary	2	1 mark for each bullet, to a maximum of 2.
	b		• The quality will improve ... • ... because the sound wave is more accurate to the original • The file size will increase ... • ... because there are more samples to store	4	1 mark for each bullet. (1 mark for identification of the effect, one mark for an explanation)
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
10		i	• The height / amplitude of the sound wave is measured • ... at regular intervals • ... and converted to binary.	2	Remember to transfer marks between (i) and (ii) if necessary... many candidates may make this point in their answer to part (ii) <u>Examiner's Comments</u> This question was poorly answered and it was obvious that many of the candidates had not studied this. Most candidates thought sampling was the same as compression, while others used their experience of downloading large media files to describe sampling as a "taster file" to determine whether the rest of the file is of acceptable quality.
		ii	• If the interval is smaller / if you sample more often you have more data to store... • ... so larger file • but the sound reproduced is closer to the original... • ...so better quality.	3	Accept the converse ie if you sample less often you have a smaller file etc... as long as the explanation is correct <u>Examiner's Comments</u> Some of those who were on the right track lost marks because their answers lacked precision. The question asked about the effect of the sample interval but some candidates interpreted this as the sample rate. As these are reciprocals of each other, the sample rate obviously gives the wrong answers, unless the candidate specifically stated that this is what they were referring to. Other candidates were even less precise with answers like "it increases the quality" without stating what change in the sampling interval increases the quality.
			Total	5	